Digital Image Processing 7:
Segmentation- Thresholding

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- Segmentation Methods

Based on two properties of gray-level image values

- Discontinuity
  - Edge detection
- Similarity
  - Thresholding
  - Region growing / splitting / merging

Contents

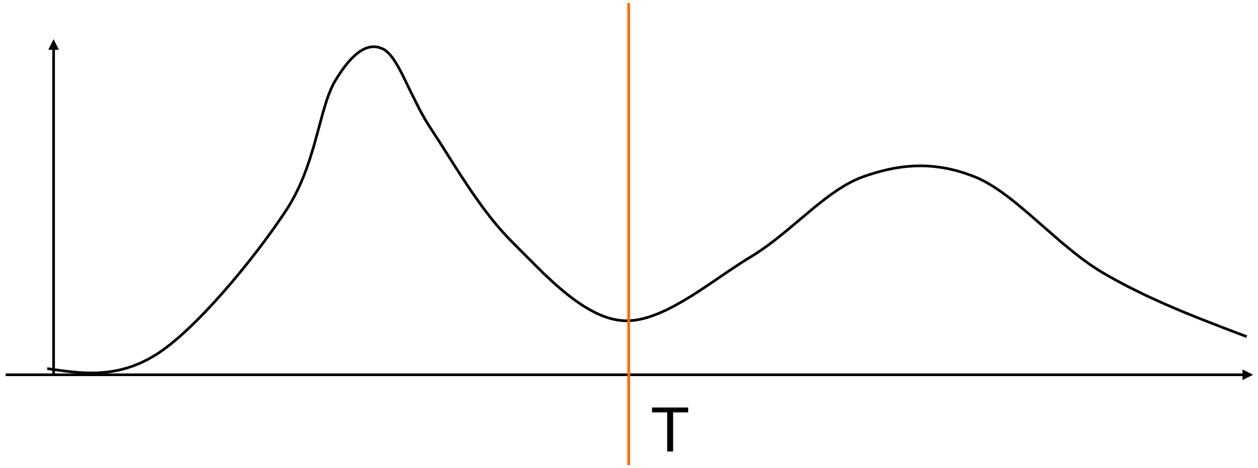
- Segmentation in terms of thresholding

- we will look at:
  - What is thresholding?
  - Global thresholding
  - Adaptive thresholding

1 What is thresholding?

- Thresholding is usually the first step in some segmentation application.
- **Single value thresholding** can be given mathematically as follows:

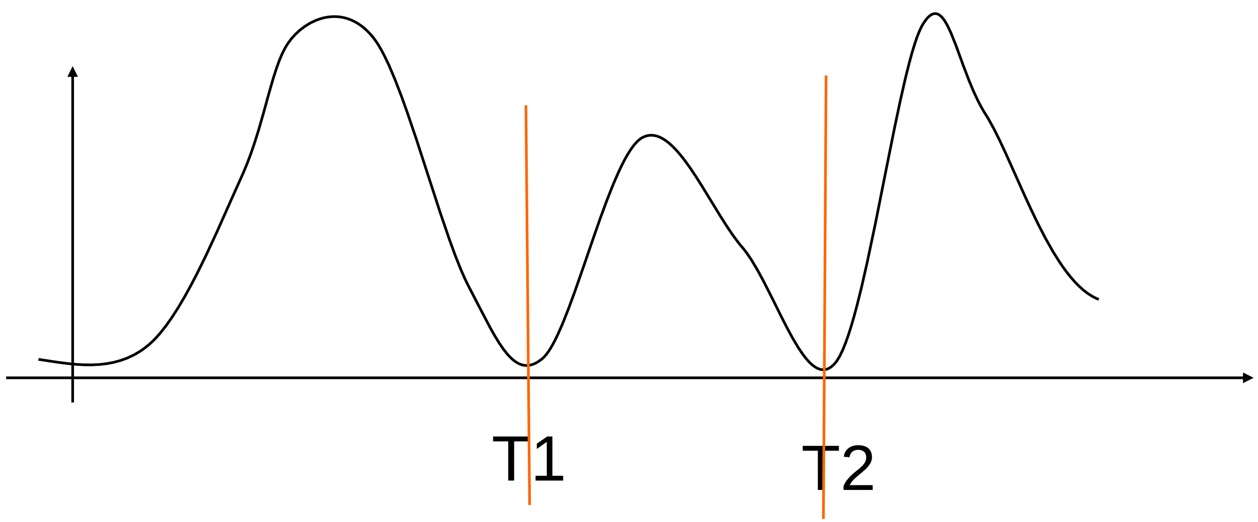
$$g(x,y)=\begin{cases}1\,if\;\; f(x,y)>T\\ 0\,if\;\; f(x,y)\le T\end{cases}$$



1 What is thresholding?

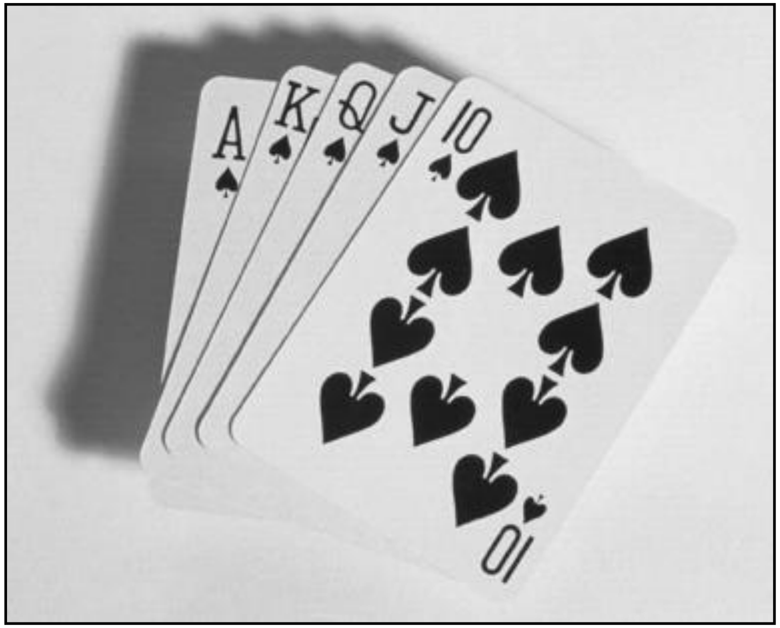
- **Multi-value thresholding** can be given mathematically as follows:

$$g(x,y)=\begin{cases} a, if \ f(x,y)>T2 \\ b, if \ f(x,y)\le T1 \\ c, if \ \mathrm{T1}\angle f(x,y)\angle T \end{cases}$$

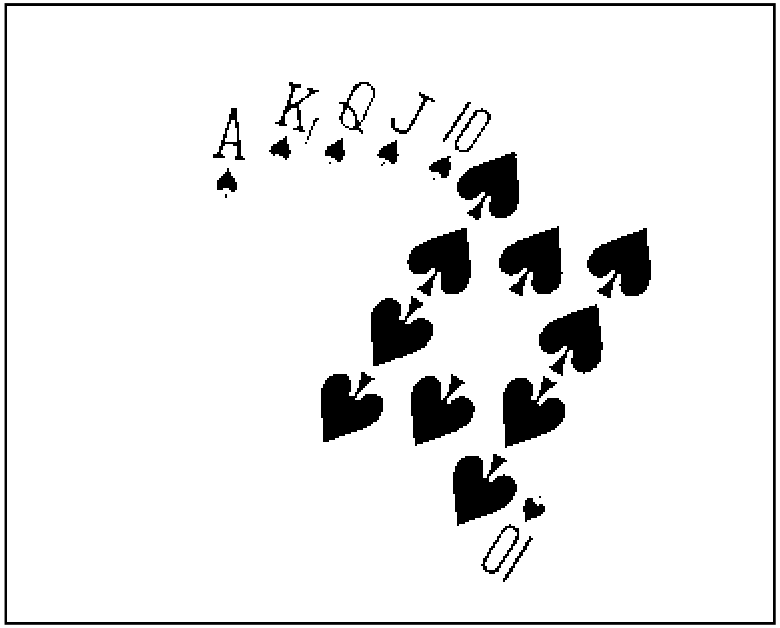


Thresholding Example

- Imagine a poker playing robot that needs to visually interpret the cards in its hand

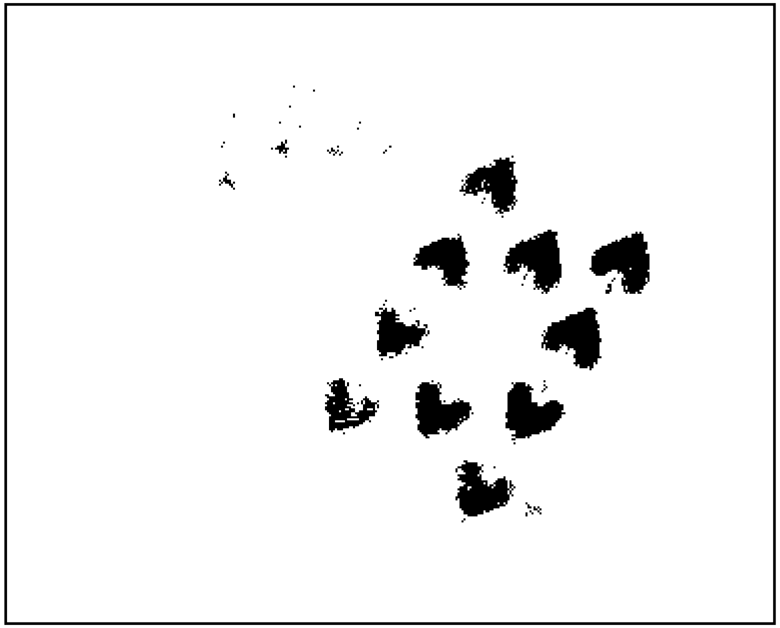


Original Image

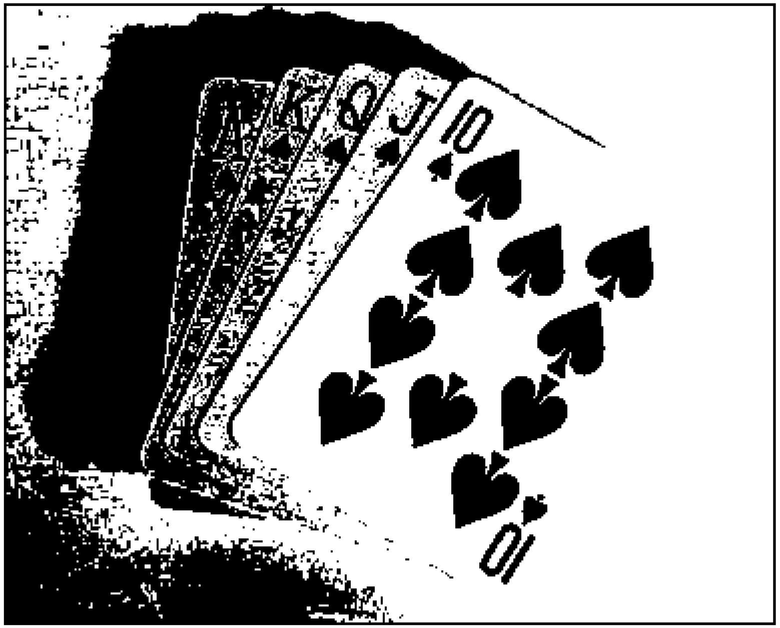


Thresholded Image

- If you get the threshold wrong the results can be disastrous



Threshold Too Low



Threshold Too High

1 What is thresholding?

- Thresholding-based segmentation requires:

a) The separation between peaks(further better?)
b) Noise content in the image (boarder as noise increase?)
c) The relative size of object and background.
d) Uniformity of the illumination source.
e) Uniformity of the reflectance properties of the image.

The width and depth of the valleys separating the histogram

2 Basic Global Thresholding

- Based on the histogram of an image
- Partition the image histogram using a single global threshold T.

- The basic global thresholding is calculated as follows:
  1. Select an initial estimation for T (typically the average grey level in the image)
  2. Segment the image using T to produce two groups of pixels: G_1 consisting of pixels with grey levels >T and G_2 consisting pixels with grey levels ≤ T
  3. Compute the average grey levels of pixels in G_1 to give μ_1 and G_2 to give μ_2

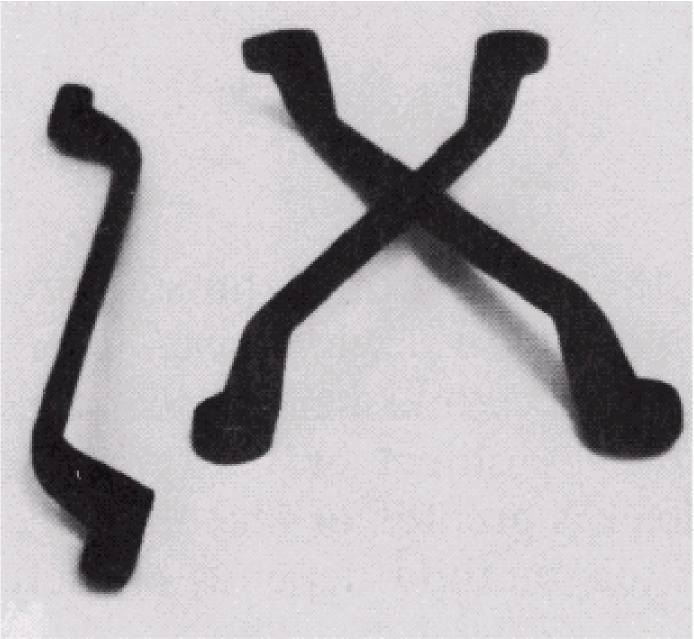
4. Compute a new threshold value:

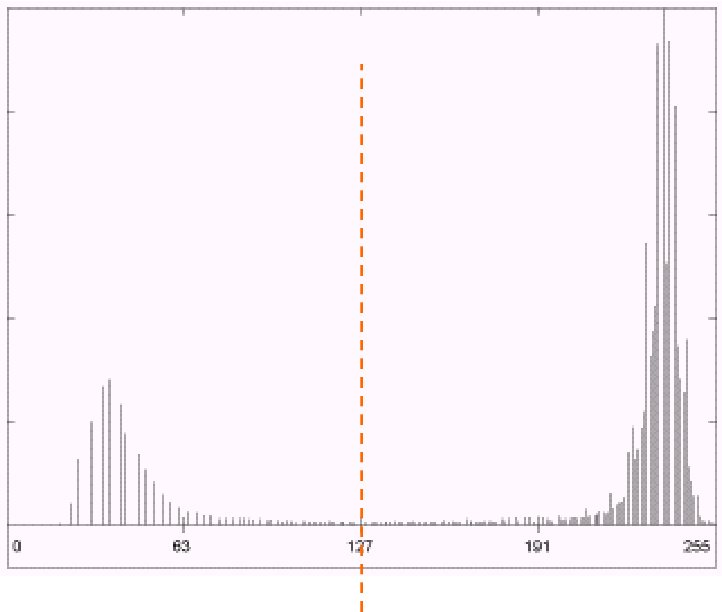
$$T = \frac{\mu_1 + \mu_2}{2}$$

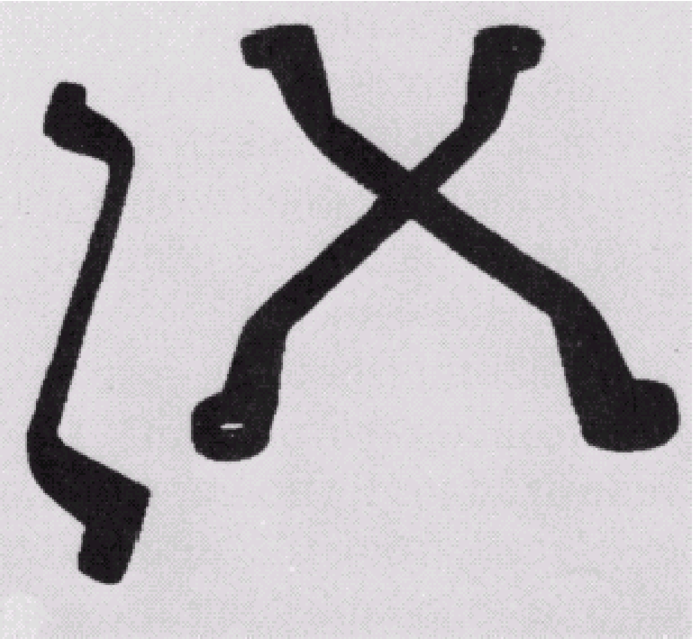
5. Repeat steps 2 – 4 until the difference in T in successive iterations is less than a predefined limit T_∞

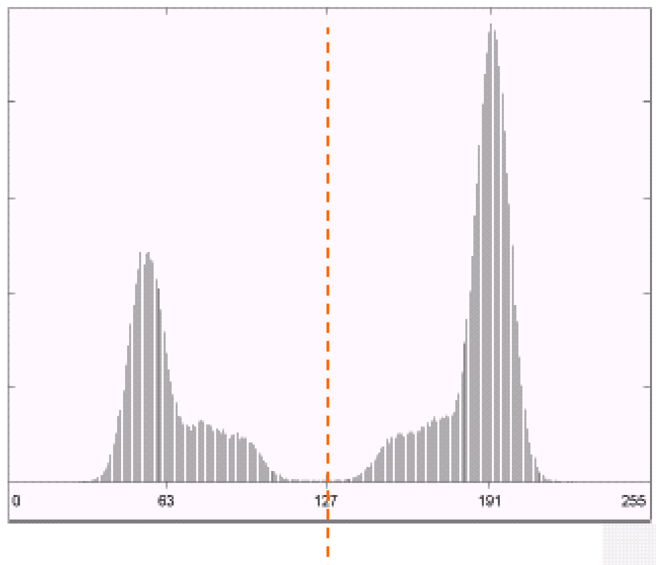
- This algorithm works very well for finding thresholds when the histogram is suitable

The success of this technique very strongly depends on how well the histogram can be partitioned！！




0
63
127
191
255




0
63
127
191
255

2 Basic Global Thresholding- Otsu’s

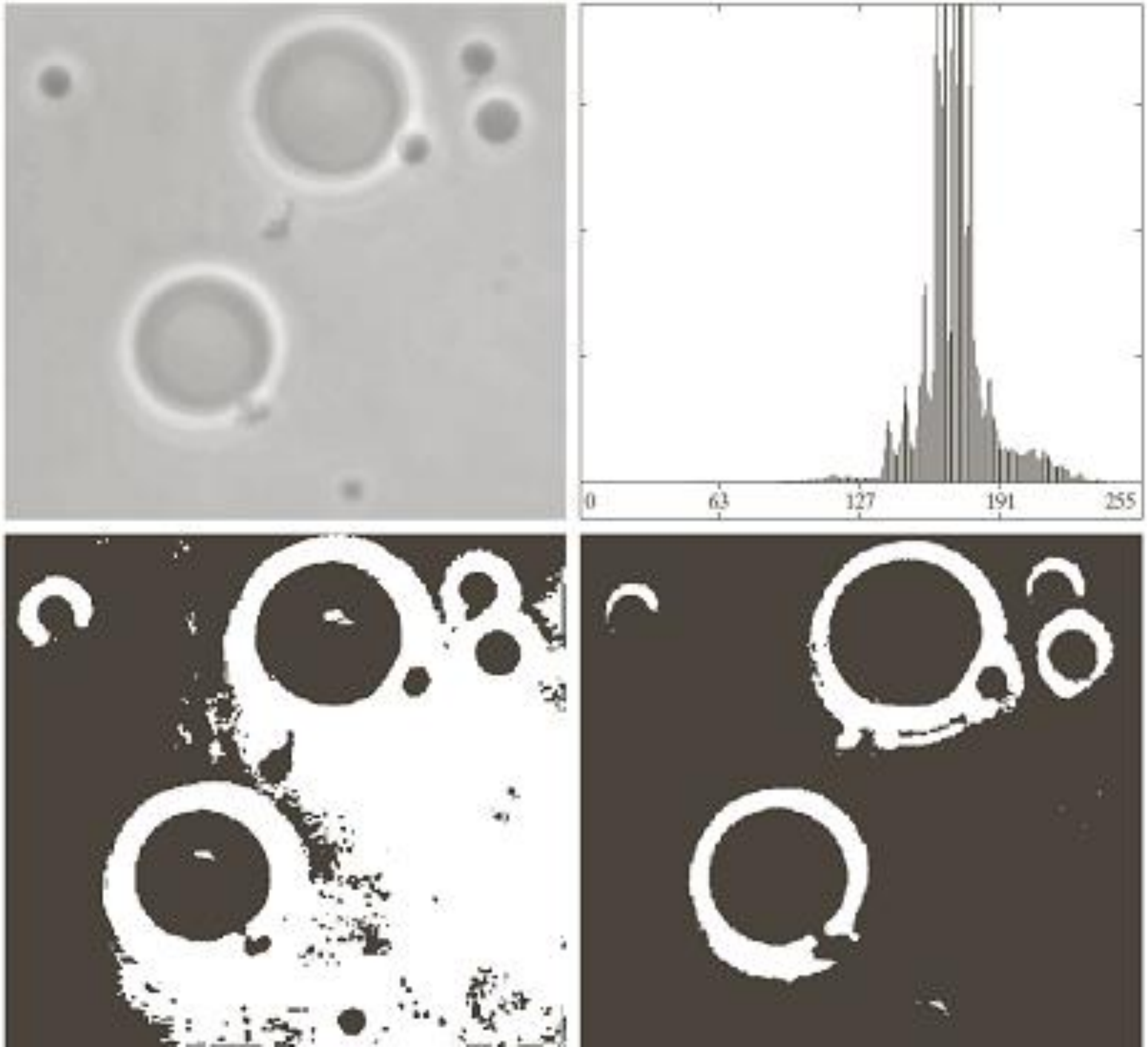
- Assumes the image histogram is bi-modal
- Maximizes between-class variance

Between class variance: $\sigma_B^2(k) = P_1(k)P_2(k)(m_1(k) - m_2(k))^2$

Probability of a pixel is assigned to Class C1: $P_1(k) = \sum_{i=0}^{k} p_i$

Probability of a pixel is assigned to Class C2: $P_2(k) = 1 - P_1(k)$

2 Basic Global Thresholding-Comparison



a b
c d

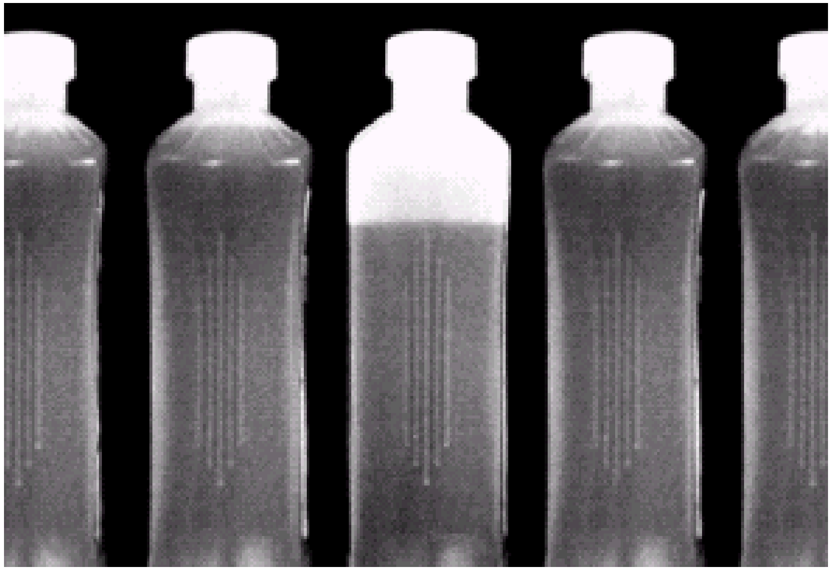
FIGURE 10.39 (a) Original image. (b) Histogram (high peaks were clipped to highlight details in the lower values). (c) Segmentation result using the basic global algorithm from Section 10.3.2. (d) Result obtained using Otsu's method. (Original image courtesy of Professor Daniel A. Hammer, the University of Pennsylvania.)

Problems With Single Value Thresholding

If we want to isolate the contents of the bottles,

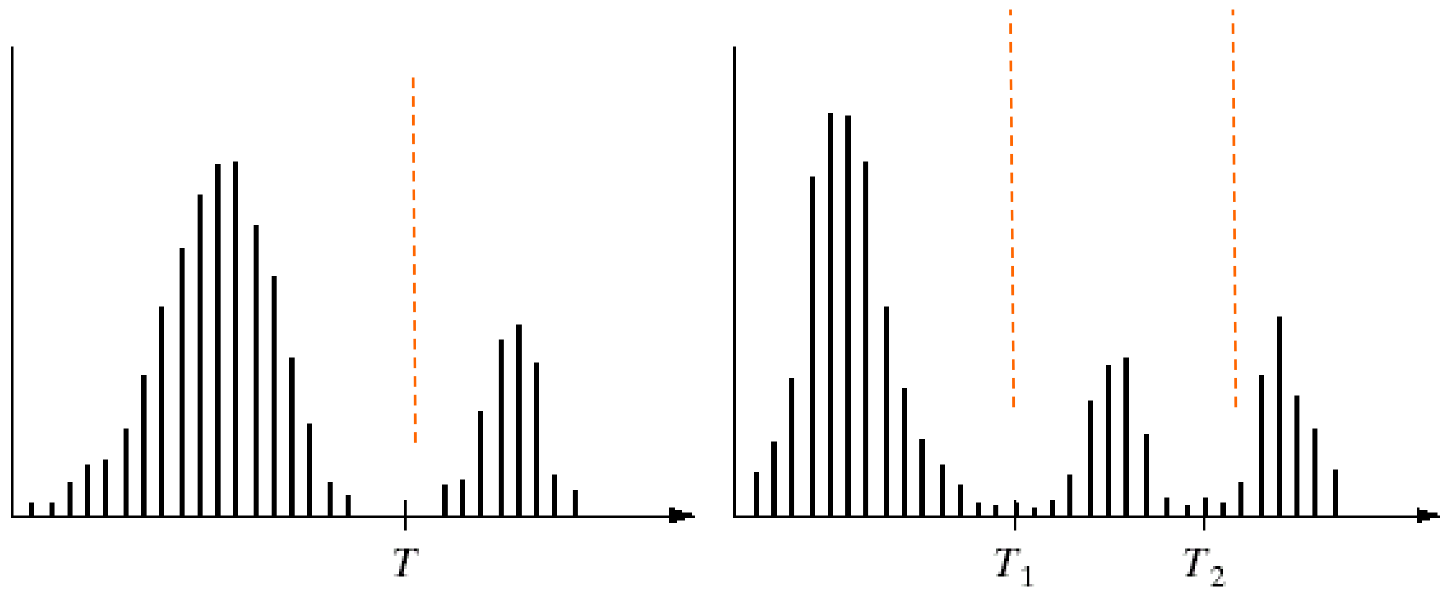
Think about what the histogram for this image would look like？

What would happen if we used a single threshold value?



Problems With Single Value Thresholding

- Single value thresholding only works for bimodal histograms
- Images with other kinds of histograms need more than a single threshold



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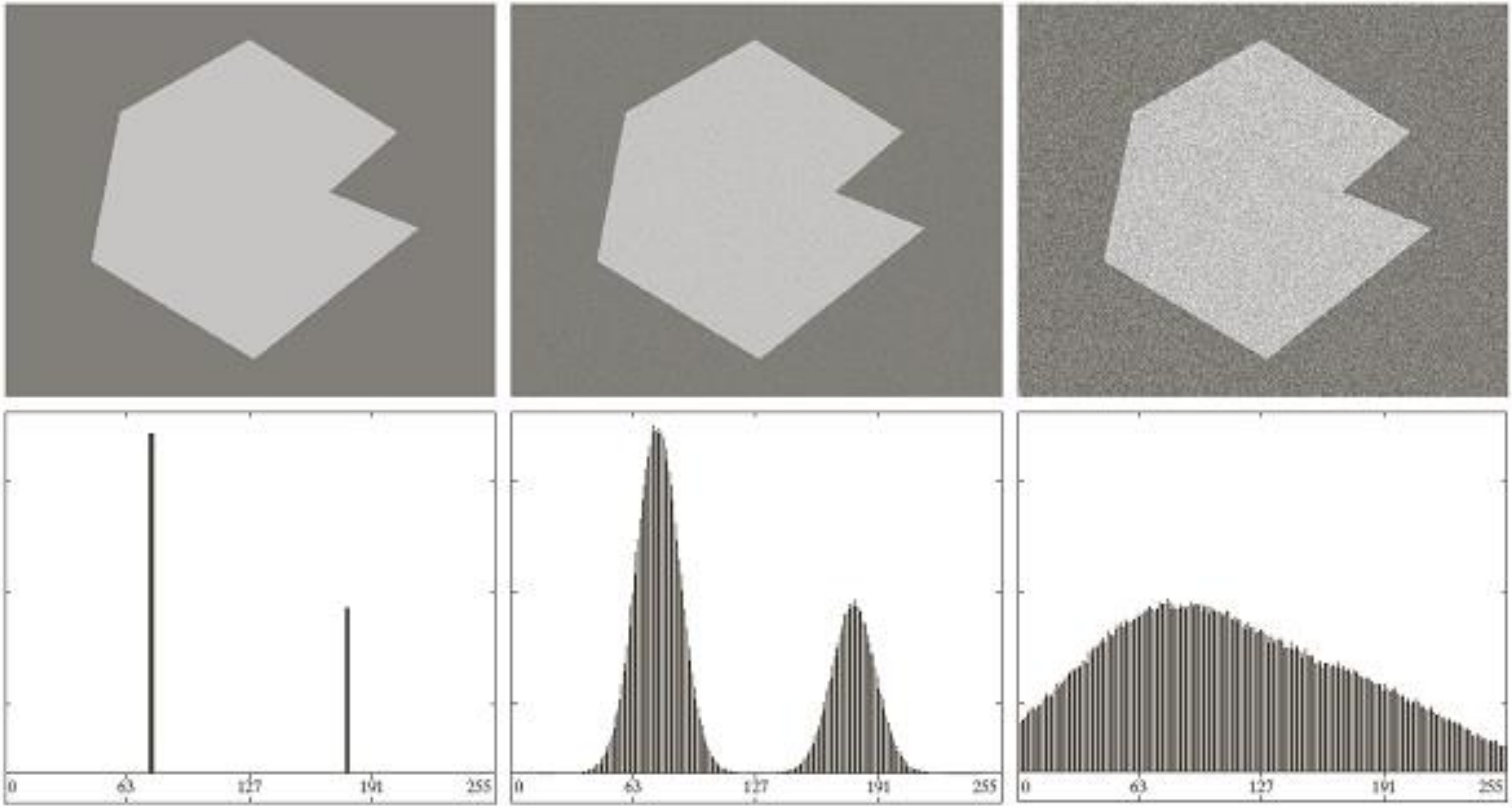
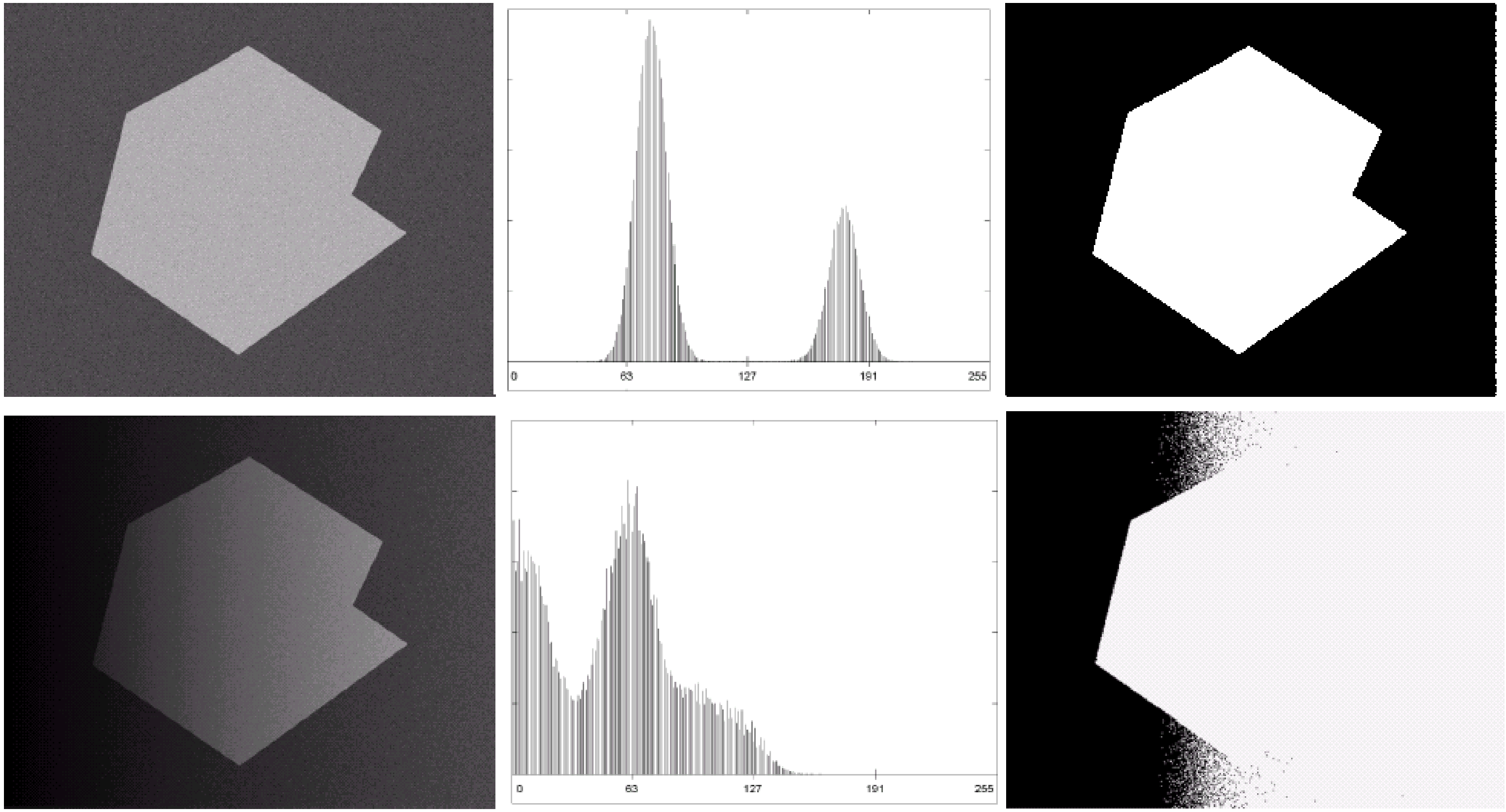
2.1 Problems With Single Value Thresholding-Nosie



- Nosie makes it’s hard to find a suitable threshold for segmenting image.

2.2 Problems With Single Value Thresholding-Illumination



- Uneven illumination can really upset a single valued thresholding scheme

2.3 Basic Global Thresholding-Smoothing

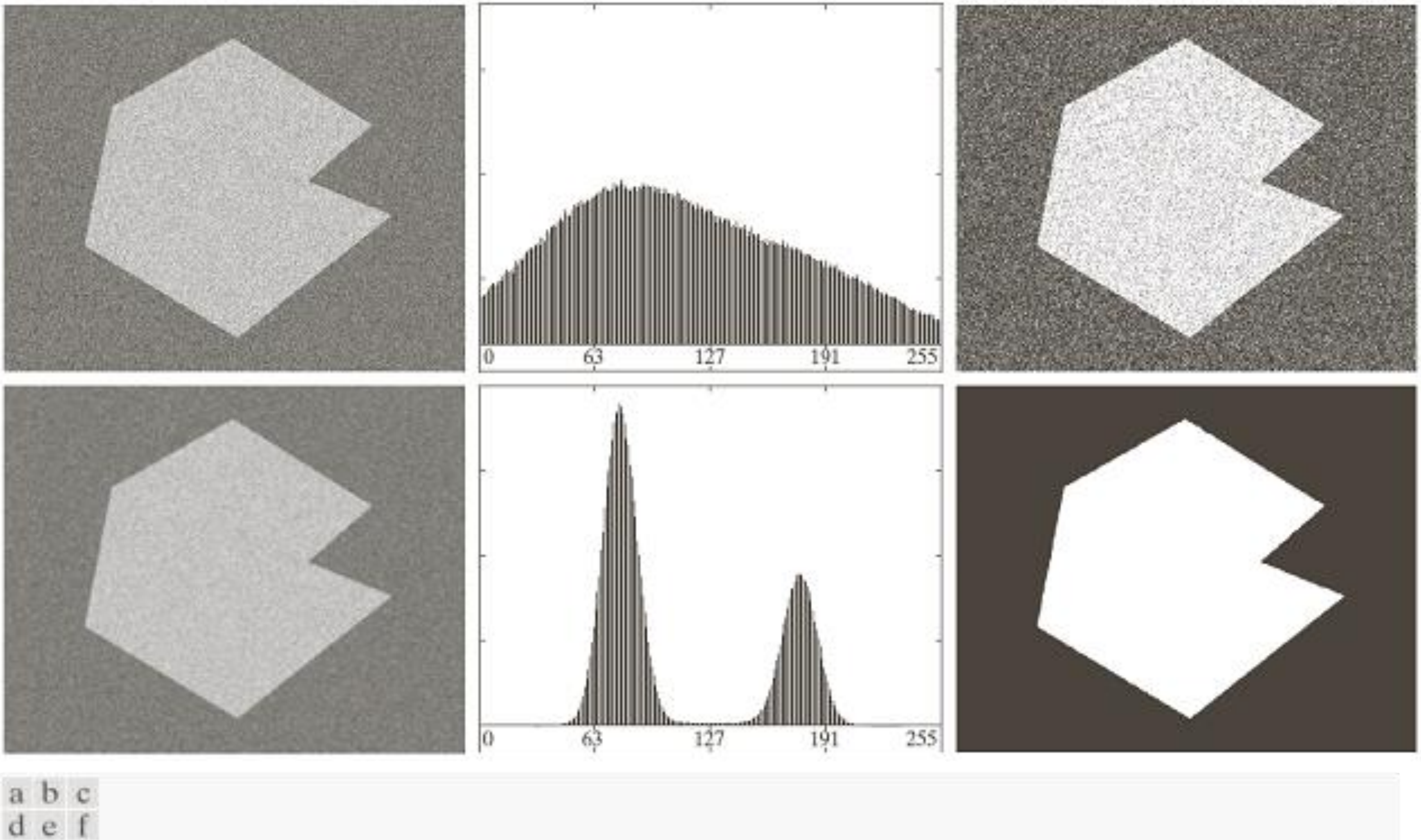


FIGURE 10.40 (a) Noisy image from Fig. 10.36 and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5 × 5 averaging mask and (e) its histogram. (f) Result of thresholding using Otsu's method.

2.4 Basic Global Thresholding-Using edges

- The chances of selecting a “good” threshold will be considerably enhanced if the histogram peaks are tall, narrow, symmetric, and separated by deep valleys.
- Approach to improve the shape of histogram
  - Only consider those pixels that lie on or near the boundary (edge)

2.4 Basic Global Thresholding-Using edges

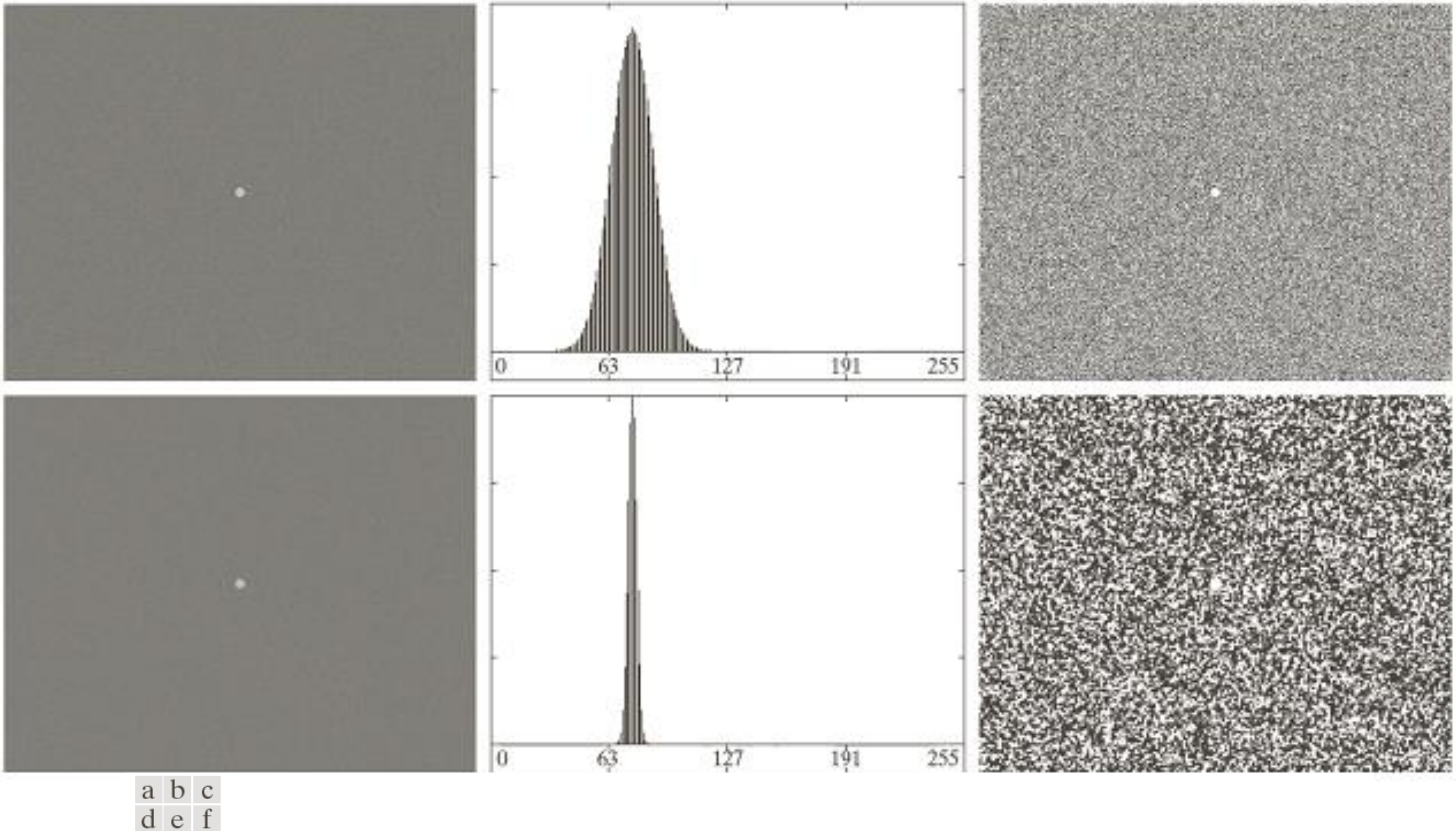
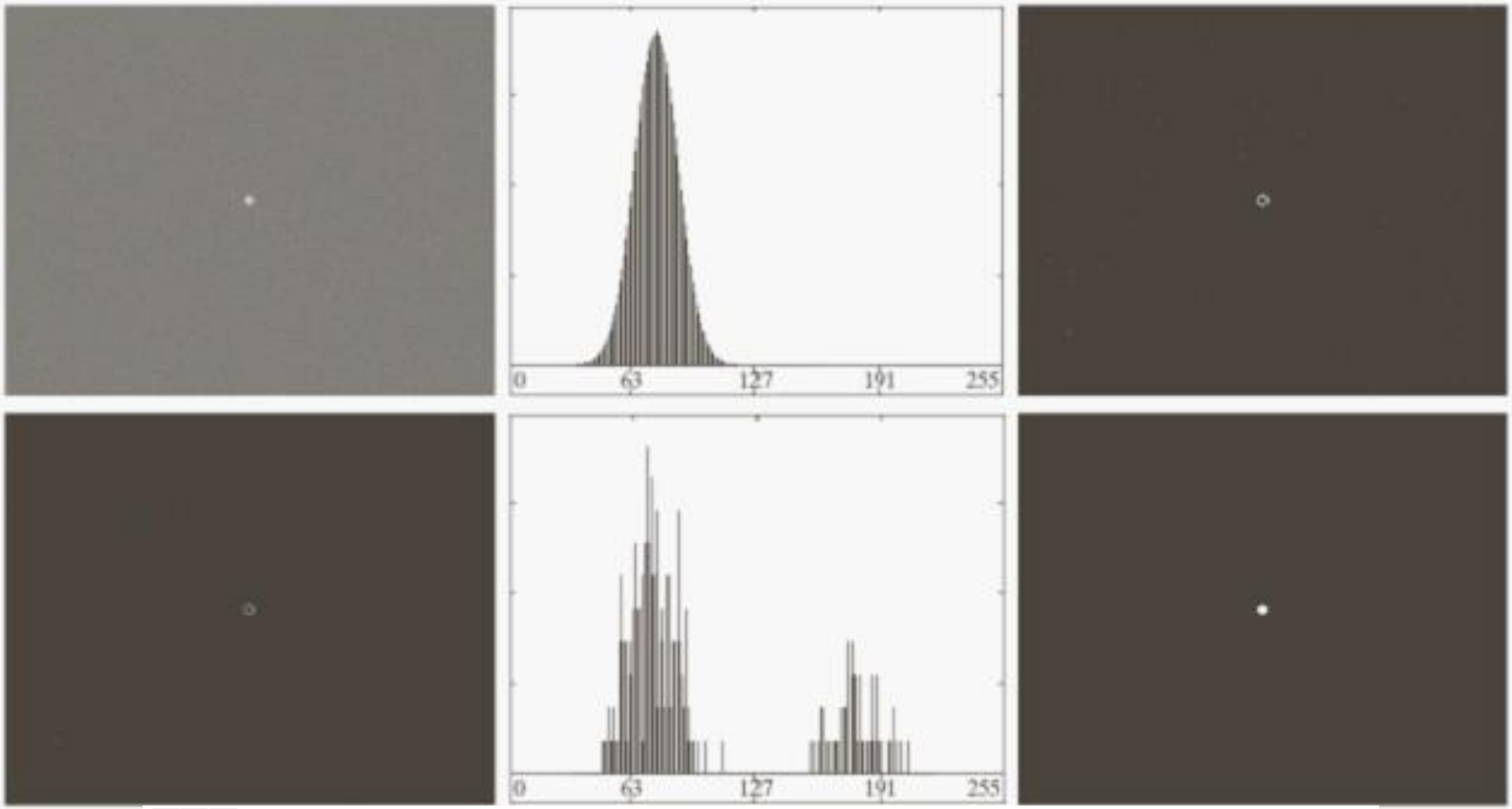


FIGURE 10.41 (a) Noisy image and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5 × 5 averaging mask and (e) its histogram. (f) Result of thresholding using Otsu's method. Thresholding failed in both cases.

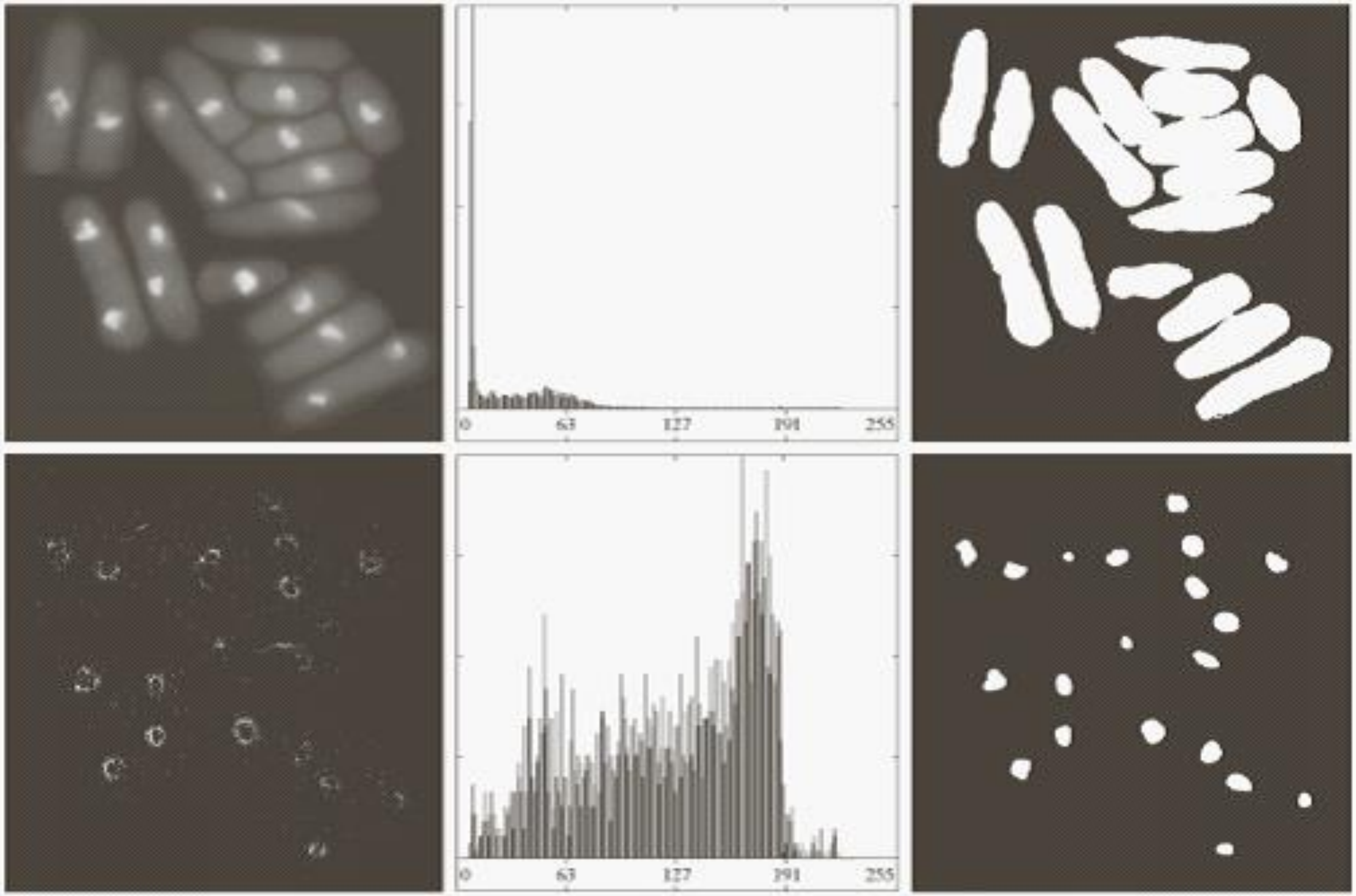
2.4 Basic Global Thresholding-Using edges



a b c
d e f

FIGURE 10.42 (a) Noisy image from Fig. 10.41(a) and (b) its histogram. (c) Gradient magnitude image thresholded at the 99.7 percentile. (d) Image formed as the product of (a) and (c). (e) Histogram of the nonzero pixels in the image in (d). (f) Result of segmenting image (a) with the Otsu threshold based on the histogram in (e). The threshold was 134, which is approximately midway between the peaks in this histogram.

2.4 Basic Global Thresholding-Using edges



a b c
d e f

FIGURE 10.43 (a) Image of yeast cells. (b) Histogram of (a). (c) Segmentation of (a) with Otsu's method using the histogram in (b). (d) Thresholded absolute Laplacian. (e) Histogram of the nonzero pixels in the product of (a) and (d). (f) Original image thresholded using Otsu's method based on the histogram in (e). (Original image courtesy of Professor Susan L. Forsburg, University of Southern California.)

3 Variable Thresholding

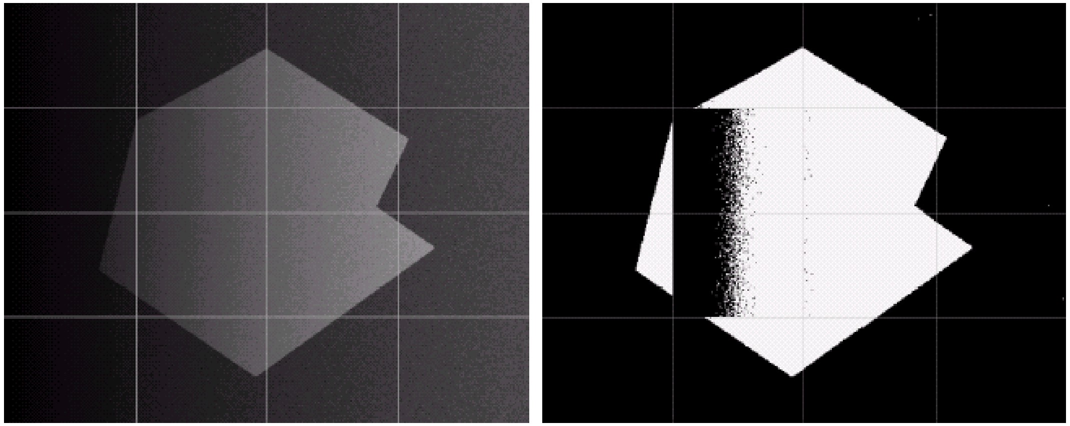
- Global thresholding sometimes doesn’t work.
- Since the threshold for each pixel depends **on its location within an image, referring to *adaptive thresholdling*.**

   a. **Image partitioning**

   b. Local image properties

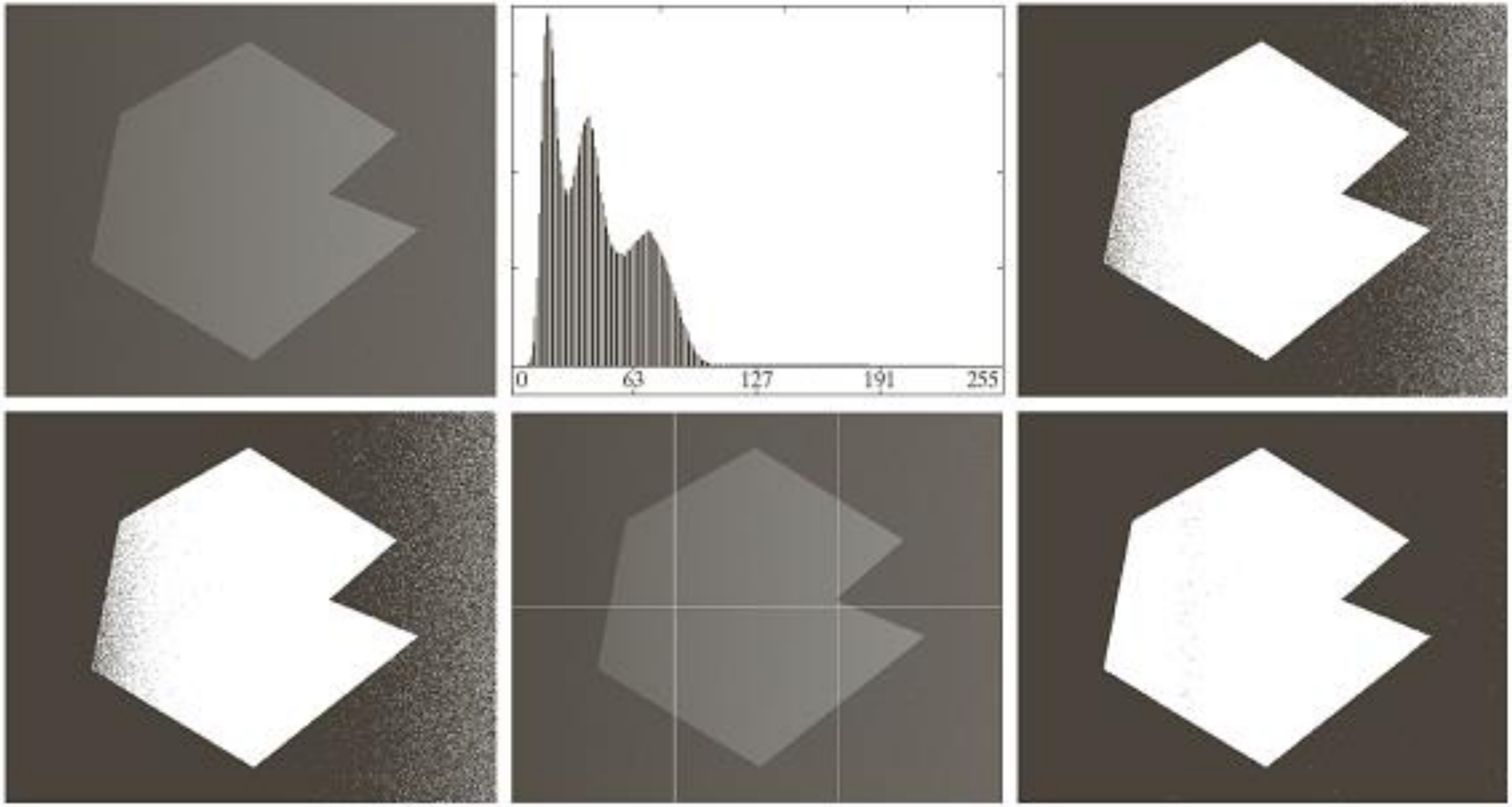
   c. Moving average

3.1 Variable Thresholding-Image partitioning



- **Method:** Subdivide an image into non-overlapping rectangles.
- **Used when:** Compensate for non-uniformities in illumination and/or reflectance.
- **Works good when:** the objects of interest and the background occupy regions of reasonably comparable size.
- **Fail when:** subdivisions containing ***only object or background.***

3.1 Variable Thresholding-Image partitioning



a b c
d e f

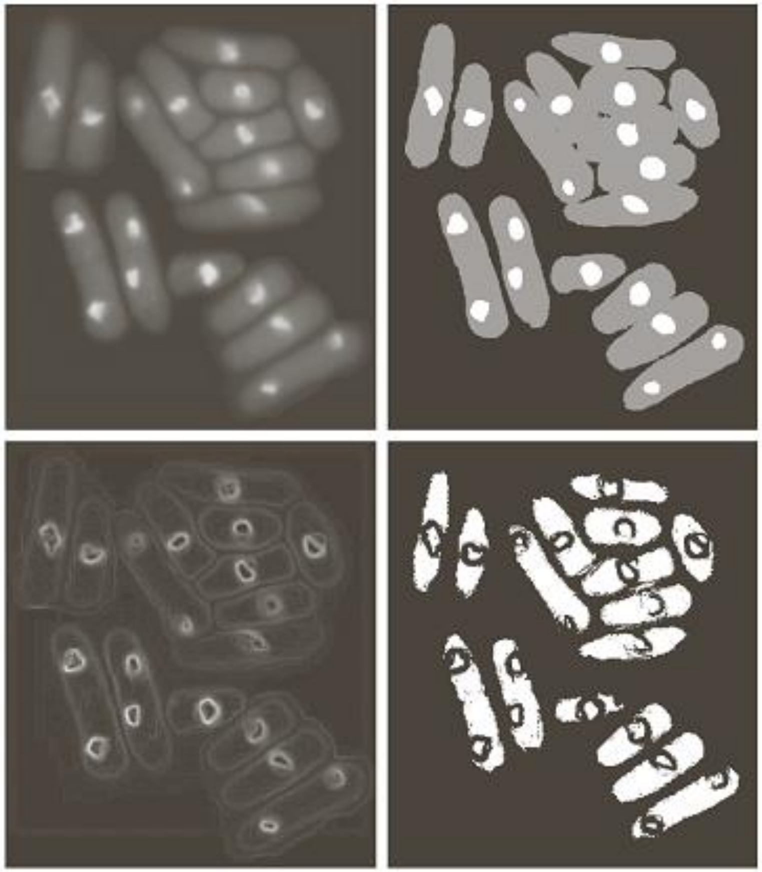
FIGURE 10.46 (a) Noisy, shaded image and (b) its histogram. (c) Segmentation of (a) using the iterative global algorithm from Section 10.3.2. (d) Result obtained using Otsu's method. (e) Image subdivided into six subimages. (f) Result of applying Otsu's method to each subimage individually.

3.2 Variable Thresholding-Local image properties

- **Method:** compute a threshold at every point based on a neighborhood
- Standard deviation σ_{xy} -local contrast
- Mean of the pixel $(m_{xy}: local; m_G: global)$- average intensity

$$T_{xy} = a\sigma_{xy} + bm_G$$

$$g(x,y) = \begin{cases} 0, & if\ f(x,y) < T_{xy} \\ 1, & if\ f(x,y) \geq T_{xy} \end{cases}$$



a b
c d

FIGURE 10.48 (a) Image from Fig. 10.43. (b) Image segmented using the dual thresholding approach discussed in Section 10.3.6. (c) Image of local standard deviations. (d) Result obtained using local thresholding.

3.3 Variable Thresholding-Moving averages

- A special case of local thresholding along scan lines of an image
- Document processing
- Zigzag pattern to reduce illumination bias

$$m(k+1) = m(k) + \frac{1}{n}(z_{k+1} - z_{k-n})$$

a b c

FIGURE 10.49 (a) Text image corrupted by spot shading. (b) Result of global thresholding using Otsu's method. (c) Result of local thresholding using moving averages.

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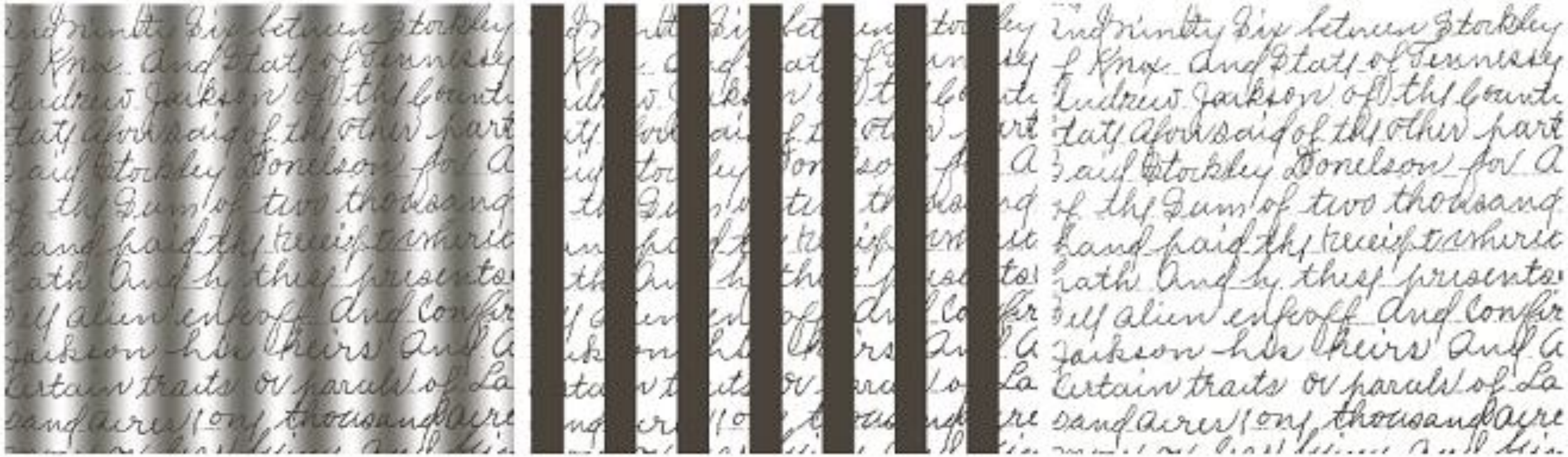
3.3 Variable Thresholding-Moving averages



a b c

FIGURE 10.50 (a) Text image corrupted by sinusoidal shading. (b) Result of global thresholding using Otsu's method. (c) Result of local thresholding using moving averages.

Objects of interests are small/thin with respect to the image size, such as images of typed/handwritten text.

4 Other methods for segmentation

Region growing-Region-based Segmentation

- Finding the regions directly
- *Definition*: a procedure that groups pixels or sub-regions into larger regions based on predefined criteria for growth.
  - Start with a set of seed point
  - Grow regions by appending to each seed those neighboring pixels have predefined properties similar to the seed.
  - Intensity, color, moment, texture, etc.

4 Other methods for segmentation

Region growing-Region-based Segmentation

- Step 1: Assume we find a good threshold, and use it to partition the regions into pure black and white
- Step 2: Use different labels to identify different objects
  - Use region growing to connect parts that should have belonged to the same region
  - This is called “Connected component analysis”
  - The regions with the same label generate one segment

Similar criteria+connectivity

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Region growing-Region-based Segmentation

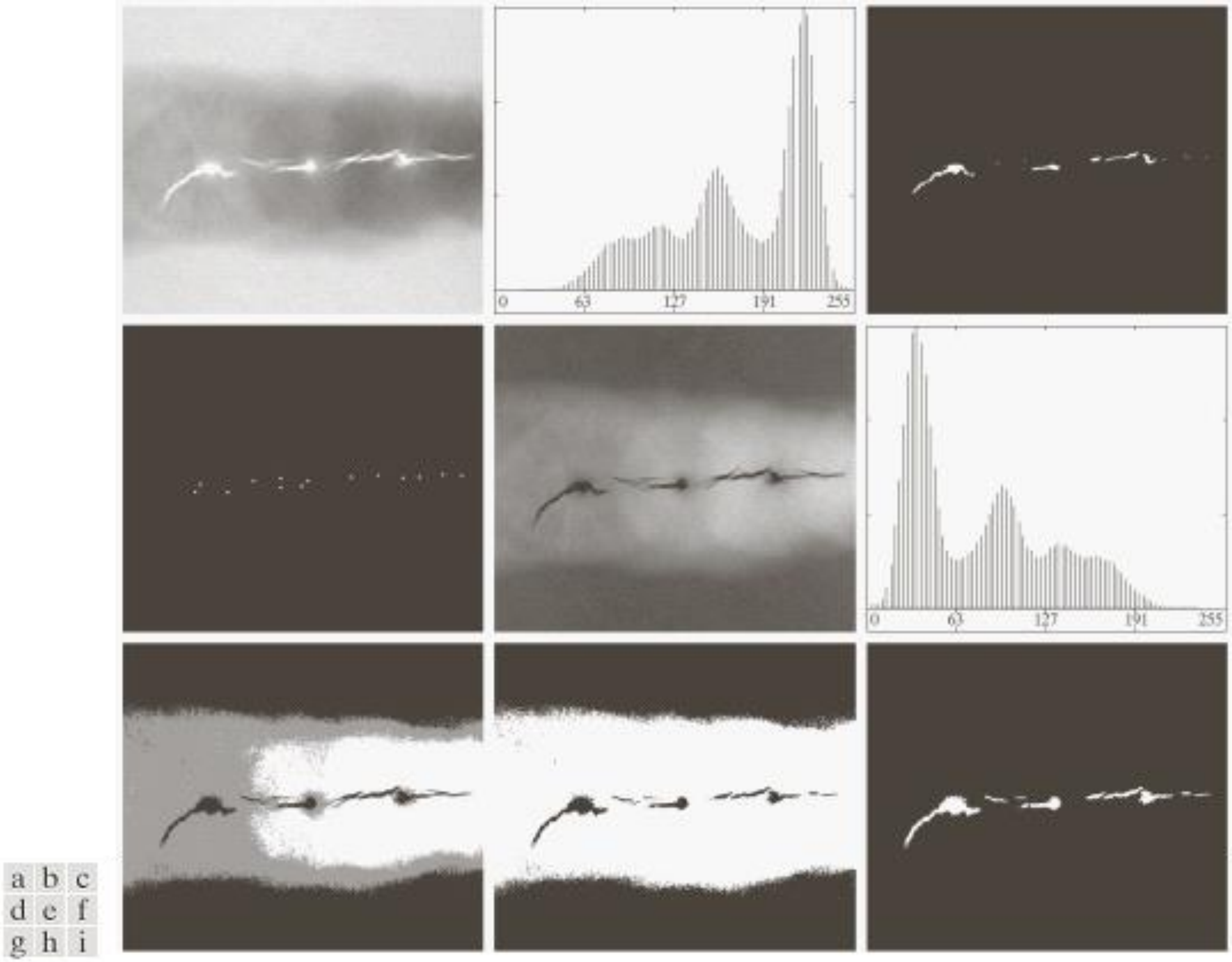
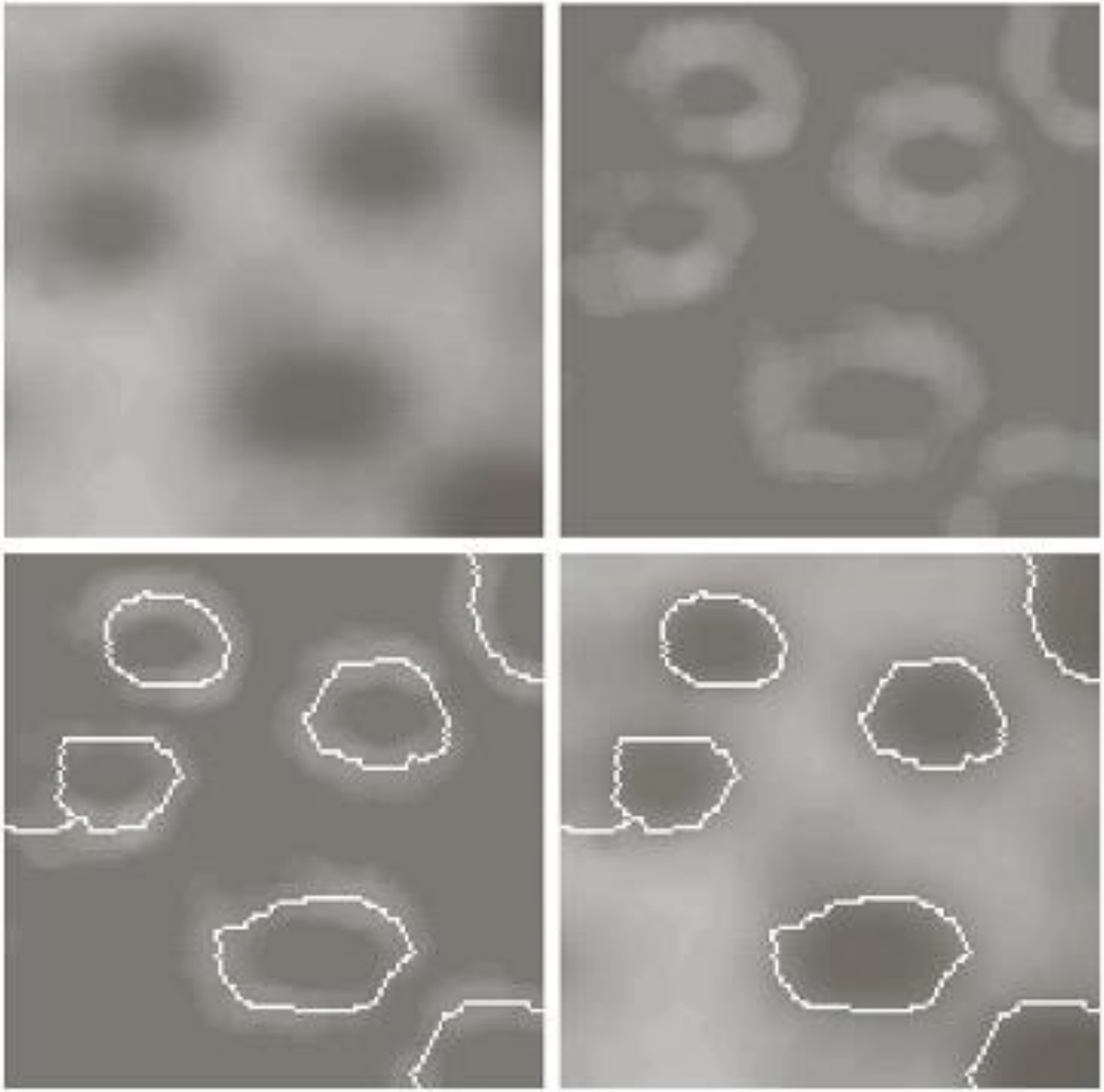


FIGURE 10.51 (a) X-ray image of a defective weld. (b) Histogram. (c) Initial seed image. (d) Final seed image (the points were enlarged for clarity). (e) Absolute value of the difference between (a) and (c). (f) Histogram of (e). (g) Difference image thresholded using dual thresholds. (h) Difference image thresholded with the smallest of the dual thresholds. (i) Segmentation result obtained by region growing. (Original image courtesy of X-TEK Systems, Ltd.)

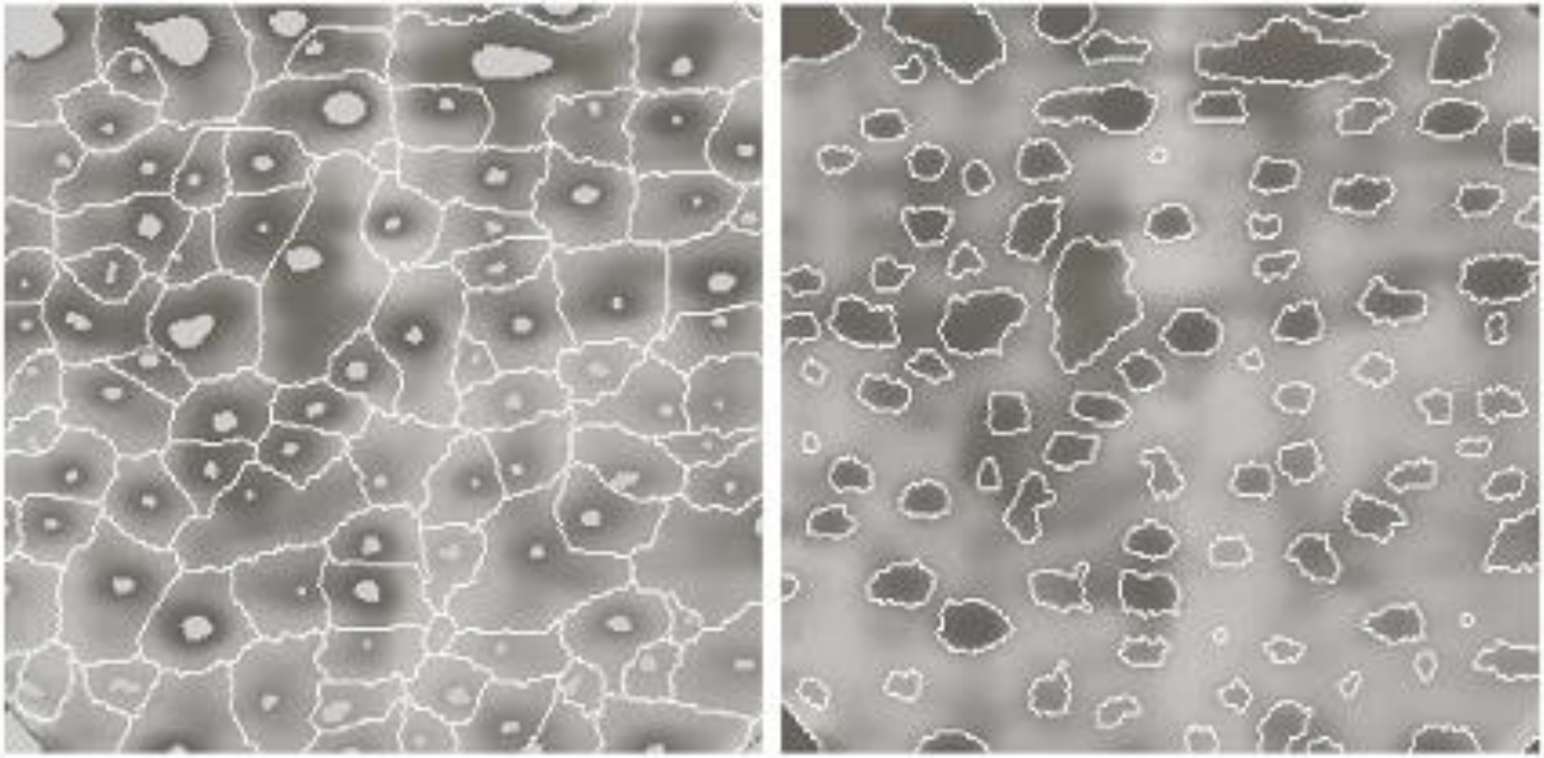
Watershed-based Segmentation



a b
c d

FIGURE 10.56 (a) Image of blobs. (b) Image gradient. (c) Watershed lines. (d) Watershed lines superimposed on original image. (Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)

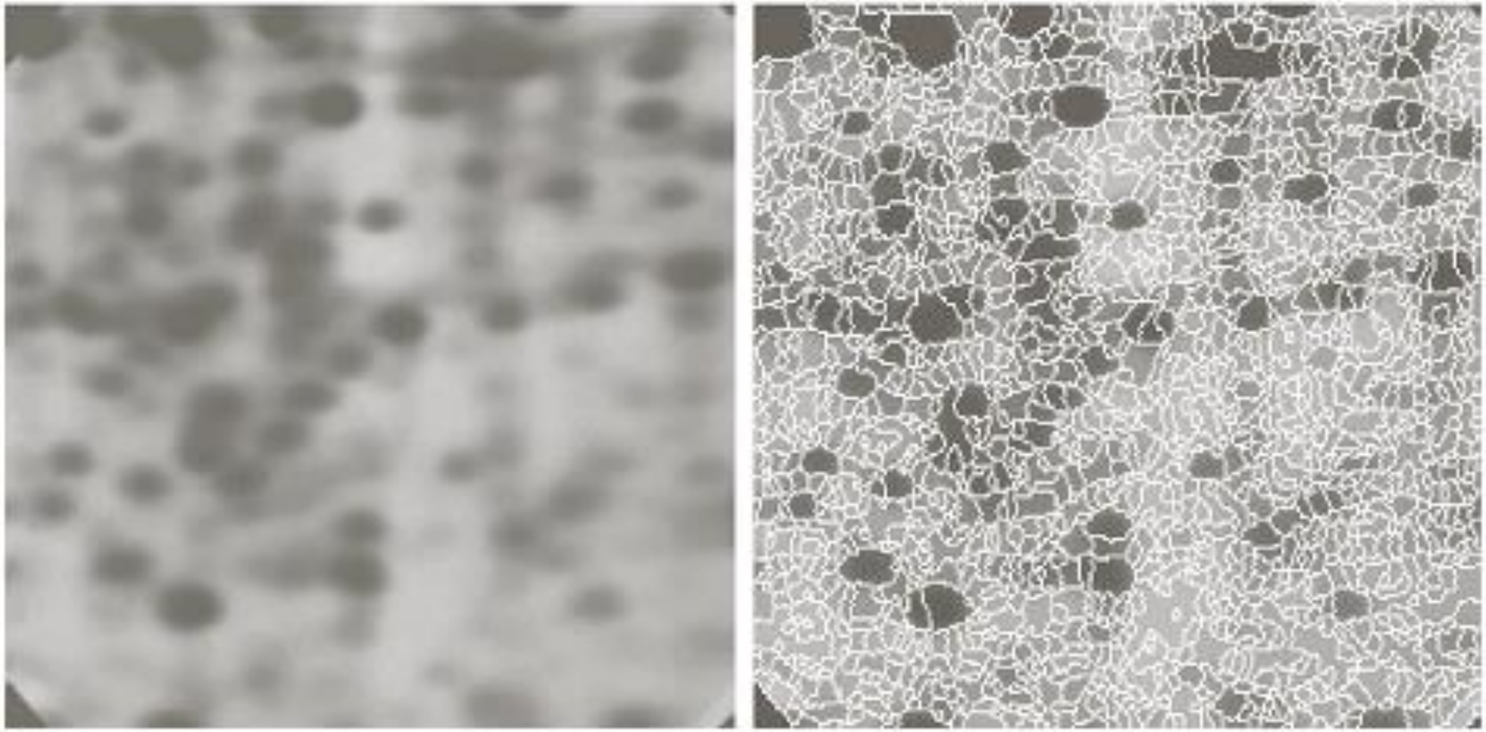
Watershed-based Segmentation



a b

FIGURE 10.58 (a) Image showing internal markers (light gray regions) and external markers (watershed lines). (b) Result of segmentation. Note the improvement over Fig. 10.47(b). (Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)

Watershed-based Segmentation



a b

FIGURE 10.57 (a) Electrophoresis image. (b) Result of applying the watershed segmentation algorithm to the gradient image. Oversegmentation is evident. (Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)

Summary

- Segmentation based on thresholding
- Basic global thresholding algorithm and its shortcomings
- A simple way to overcome some of these limitations using adaptive thresholding